

Problem 3

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Problem 1 Evaluate the following limits.

$$(a) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos(x)}{x}$$

Explanation. Apply the Quotient Law and use continuity of the numerator and denominator.

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos(x)}{x} = \frac{0}{\frac{\pi}{2}} = 0$$

$$(b) \lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 4}$$

Explanation. Apply the Quotient Law and use continuity.

$$\lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 4} = \frac{\sqrt{25 - (3)^2} - 4}{3 - 4} = \frac{4 - 4}{-1} = 0$$

$$(c) \lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 3}$$

Explanation. Since $\lim_{x \rightarrow 3} (\sqrt{25 - x^2} - 4) = 0$ and $\lim_{x \rightarrow 3} x - 3 = 0$, this limit has form $\frac{0}{0}$.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 3} &= \lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 3} \cdot \frac{\sqrt{25 - x^2} + 4}{\sqrt{25 - x^2} + 4} = \\ &= \lim_{x \rightarrow 3} \frac{25 - x^2 - 16}{(x - 3)(\sqrt{25 - x^2} + 4)} = \lim_{x \rightarrow 3} \frac{9 - x^2}{(x - 3)(\sqrt{25 - x^2} + 4)} = \lim_{x \rightarrow 3} \frac{(3 + x)(3 - x)}{(x - 3)(\sqrt{25 - x^2} + 4)} = \\ &= \lim_{x \rightarrow 3} \frac{-(3 + x)}{\sqrt{25 - x^2} + 4} = \frac{-6}{\sqrt{25 - 9} + 4} = \frac{-6}{\sqrt{16} + 4} = \frac{-6}{8} = \frac{-3}{4} \end{aligned}$$